Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r,\theta,t) = \frac{p_0 \cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$

$$\boldsymbol{A}(r,\theta,t) = -\frac{\mu_0 p_0 \omega}{4\pi r}\sin\left[\omega\left(t-\frac{r}{c}\right)\right]\hat{\boldsymbol{z}}$$

satisfy the Lorenz gauge condition.

Solution:

□

Problem 11.2

Problem 11.3

Problem 11.4

Problem 11.5

Problem 11.6

Problem 11.7

Problem 11.8

Problem 11.9

Problem 11.10

Problem 11.11

Problem 11.12

Problem 11.13

Problem 11.14

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Problem 11.32

Problem 11.33

Problem 11.34

Problem 11.35

Problem 11.36